

Full Length Article

Integrative approach of machine learning and symbolic regression for stability prediction of multicomponent perovskite oxides and high-throughput screening

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ARTICLE INFO

Keywords:

Symbolic regression
Machine learning
High-throughput
Perovskite oxides

ABSTRACT

This work unfolds a robust and interpretable strategy for evaluating the stability and potential photovoltaic application of 6526 multicomponent perovskite oxides, employing a synergetic methodology that intertwines advanced machine learning (ML) algorithms and symbolic regression based on genetic programming. Initially, ML algorithms, namely XGBoost, LightGBM, and random forest, were harnessed, with elemental oxidation state and electronegativity serving as input features, achieving R^2 values of 0.98, 0.98, and 0.74, respectively, on the test set for predicting the formation enthalpy, a criterion for perovskite stability. Despite the amplified interpretability offered by SHAP analysis, the inherent “black-box” nature of ML obfuscates a transparent understanding of intrinsic relations between input features and performance. To surmount this, symbolic regression introduced not only elucidates a clear functional relationship between input features and perovskite stability but also achieves a commendable R^2 of 0.79 on the test set. Subsequent high-throughput screening, based on perovskite stability ranking, designated the top 500 stable perovskites for band gap calculation using the PBE functional, wherein $\text{DyNdHf}_2\text{O}_6$, $\text{CeEuAl}_2\text{O}_6$, and $\text{CeSrAl}_2\text{O}_6$ emerged as potential candidates for photovoltaic applications and were subjected to further electronic structure simulations employing the HSE06 functional, encompassing density of states, band structure, charge density, and optical absorption spectra. Ultimately, $\text{CeEuAl}_2\text{O}_6$, boasting an optical direct bandgap of 2.31 eV and minimal electron-hole wavefunction overlap, stands out as the prime choice for photovoltaic materials. This research not only pioneers the exploration of enhancing the interpretability of ML but also propels theoretical guidance for the evolution of photovoltaic cells by bridging predictive modeling with high-throughput screening.

1. Introduction

In recent years, perovskite oxides have achieved notable advancements in catalysts [1–4] and photovoltaic cells [5–7] domains. Given its chemical formula of $\text{AA}'\text{BB}'\text{O}_6$, which offers manifold possibilities for element substitution, the potential compounds of perovskite oxides span a vast space. This makes the thorough exploration of its chemical space a formidable challenge. Machine learning (ML) offers a promising avenue to enhance exploration efficiency. For instance, XGBoost [8] has demonstrated capabilities in predicting the high dielectric breakdown strength of perovskite materials for capacitors [9] and interactions between ions and low-dimensional halide perovskites [10]. LightGBM [11], on the other hand, has been effective in predicting perovskite formation energies [12] and dielectric constants [13]. Meanwhile,

random forest [14] has shown progress in anticipating the impact of cation–anion ratios and annealing temperatures on the power conversion efficiency of perovskites [15].

XGBoost [8], standing for eXtreme Gradient Boosting, pivots around the gradient boosting framework. What sets it apart is its regularization boosting. LightGBM [11], an acronym for Light Gradient Boosting Machine, also grounds in the gradient boosting framework but introduces two pivotal enhancements: Gradient-based One-Side Sampling and Exclusive Feature Bundling. These innovations enable LightGBM to exhibit faster training speeds and better handling of imbalanced data sets. In contrast, random forest [14] operates by constructing numerous decision trees during its training phase and outputs the average predictions across these trees for regression tasks. Though a robust algorithm, it lacks the fine-tuning optimization strategies employed by

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<https://doi.org/10.1016/j.compmatsci.2024.112889>

Received 2 December 2023; Accepted 16 February 2024

Available online 19 February 2024

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XGBoost and LightGBM. The latter two, with their adaptive learning rates and ability to manage imbalanced data sets, outperform in predicting perovskite materials [16,17]. However, a prevailing concern with these ML models is their “black-box” nature, leading to a lack of transparency in predictions [18]. To address this, SHAP [19] (SHapley Additive exPlanations) analysis has been employed to decipher the relationships between features and performance in these ML models. Yet, it remains challenging to fully understand the intrinsic relationship between the model’s inputs and outputs.

To uncover the intrinsic mathematical relationships between perovskite input features and their characterization indicators, the incorporation of symbolic regression [20] based on genetic programming emerges as a judicious choice. Symbolic regression begins with a set of random mathematical formulas and employs evolutionary computational methods to iteratively generate, evaluate, and evolve mathematical expressions to optimally fit the data. Formulas adept at representing the data are more likely to be passed on to subsequent generations. Through processes akin to biological crossover and mutation, new formulas are birthed, ensuring diversity and adaptability within the population. This iterative procedure refines the pool of formulas until termination criteria are met. When juxtaposed with traditional ML, symbolic regression inherently furnishes interpretable models. The resulting mathematical expressions can be directly analyzed and comprehended, facilitating elucidation and understanding of nonlinear and intricate relationships, thus providing theoretical guidance for experiments. For instance, by taking oxidation state, ionic radii, and electronegativity as input features and employing symbolic regression, one can establish connections between these inputs and critical perovskite parameters such as formation energy [21], novel tolerance factor [22], octahedral tilting [23], oxygen vacancy defect formation energy [24], ionic mobility [25], and band gap [26]. This aids in guiding the design of high-performance perovskites.

In this study, we built upon the multicomponent perovskite oxide dataset created by Musgrave and colleagues [27], selecting 6526 compositions with chemical formulas of either $A_2BB'O_6$ or $AA'B_2O_6$. The formation enthalpy of these perovskites served as our key parameter, with initial consideration given to oxidation state, ionic radii, and electronegativity as input features. The dataset was randomly partitioned into training and testing sets with a 70:30 ratio. Both ML and symbolic regression hyperparameter searches were conducted using ten-fold cross-validation. As illustrated in Table S1, various input feature combinations were evaluated using the XGBoost algorithm on the testing set. The results highlighted that the combination of oxidation state and electronegativity yielded the best performance, achieving a coefficient of determination (R^2) of 0.98. Subsequent, more comprehensive ML and symbolic regression models were thus trained using these two features. The performance of LightGBM was comparable to XGBoost, both outperforming the random forest algorithm. To enhance the interpretability of the ML models, SHAP analysis was utilized to identify which specific structural positions’ oxidation state or electronegativity exert a pronounced influence on the formation enthalpy. Additionally, a genetic programming-based symbolic regression algorithm was employed to derive an explicit function connecting input features of oxidation state and electronegativity with formation enthalpy, addressing the interpretability limitations of black-box ML models. This achieved an R^2 of 0.79 on the test set, surpassing the random forest algorithm by 0.05. To explore the potential applications of multicomponent perovskite oxides in photovoltaics, we selected the top 500 perovskite materials based on stability rankings, specifically those with the largest absolute negative formation enthalpy values. Band calculations under the Perdew–Burke–Ernzerhof (PBE) [28] functional were performed on these materials, with three perovskites closest to the Shockley–Queisser limit [29] being identified: $DyNdHf_2O_6$, $CeEuAl_2O_6$, and $CeSrAl_2O_6$. Subsequent simulations for these three compounds, under the Heyd–Scuseria–Ernzerhof (HSE06) [30] functional, covered the density of states, band structure, charge

density, and optical absorption spectra. Notably, $CeEuAl_2O_6$, with its suitable band gap and minimal electron-hole wavefunction overlap, emerges as a promising contender for next-generation photovoltaic materials. Further details are available in the supporting information.

2. Computational methods

In this study, calculations were carried out at the level of density functional theory (DFT) using the vienna ab initio simulation package (VASP) based on a plane-wave basis set [31–34]. The projector augmented wave (PAW) method [35,36] was employed for the pseudopotentials. Exchange-correlation potentials were described using the generalized gradient approximation (GGA) in the form of the PBE functional [28] and the HSE06 functional with 25 % Hartree–Fock exchange [30]. The Grimme DFT-D3 correction was incorporated to account for long-range van der Waals interactions [37]. The plane-wave cutoff energy was set to 500 eV. Total energy was considered converged once the difference between two consecutive self-consistent electronic cycles was less than 1×10^{-5} eV, and the force convergence threshold was set to 0.01 eV/Å. The structural parameters of the perovskite supercells were fully optimized using a $4 \times 4 \times 4$ Monkhorst-Pack k-point grid [38]. Subsequently, the projected density of states (PDOS) of the perovskites was computed using a $6 \times 6 \times 6$ Monkhorst-Pack k-point grid, and an 11-grid point between the high symmetry points in the k-path was employed for the band structure calculation.

Fig. 1 depicts the workflow for predicting the stability of perovskites through machine learning and symbolic regression, as well as for high-throughput screening of their photovoltaic application potential. A dataset of 6526 $AA'BB'O_6$ multinary perovskite oxides was compiled, utilizing oxidation states and electronegativity as input features, and the corresponding formation enthalpy as the stability index. This data was randomly split into training and testing sets in a 70:30 ratio. Hyperparameters for both machine learning and symbolic regression models were searched using ten-fold cross-validation. For XGBoost and LightGBM, the maximum number of iterations was set to 5000, with early stopping employed to determine the optimal number of iterations and prevent overfitting. For the random forest, the number of trees was set at 500. Symbolic regression was implemented using the PySR code [39], and the optimal hyperparameter settings were detailed in Table S2. Subsequently, predictive performance was assessed on the test set. The accuracy of the model was evaluated based on mean absolute error (MAE), root mean square error (RMSE), and R^2 , defined as follows:

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i)^2}{n}}$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - x_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Here, x_i and y_i denote the computed and predicted values, respectively, while $\bar{x}$ represents the average of computed values. Following this, the top 500 stable perovskites, ranked by formation enthalpy, were selected for bandgap calculations under the PBE functional. Three perovskites with appropriate bandgaps were chosen for more detailed electronic structure simulations, aiming to identify potential high-performance photovoltaic materials.

3. Results and discussion

We initially evaluated the performance of various ML models on a dataset consisting of formation enthalpies for 6526 perovskites, as depicted in Fig. 2. Under the early stopping criterion, optimal iteration counts for XGBoost and LightGBM were identified as 1847 and 1248,

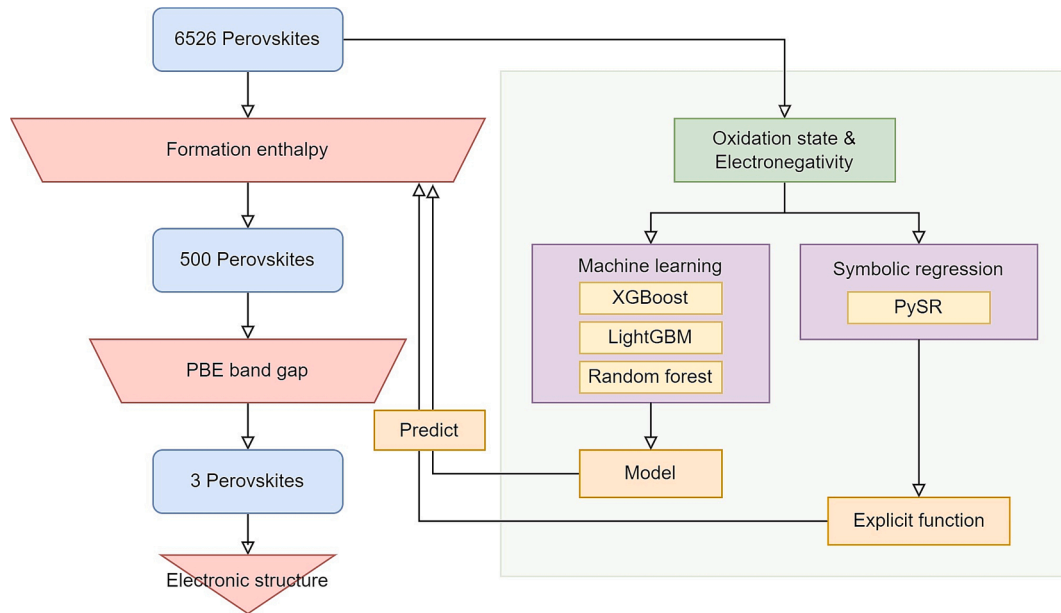


Fig. 1. Flowchart of the machine learning and symbolic regression approach for predicting perovskite stability and high-throughput screening process.

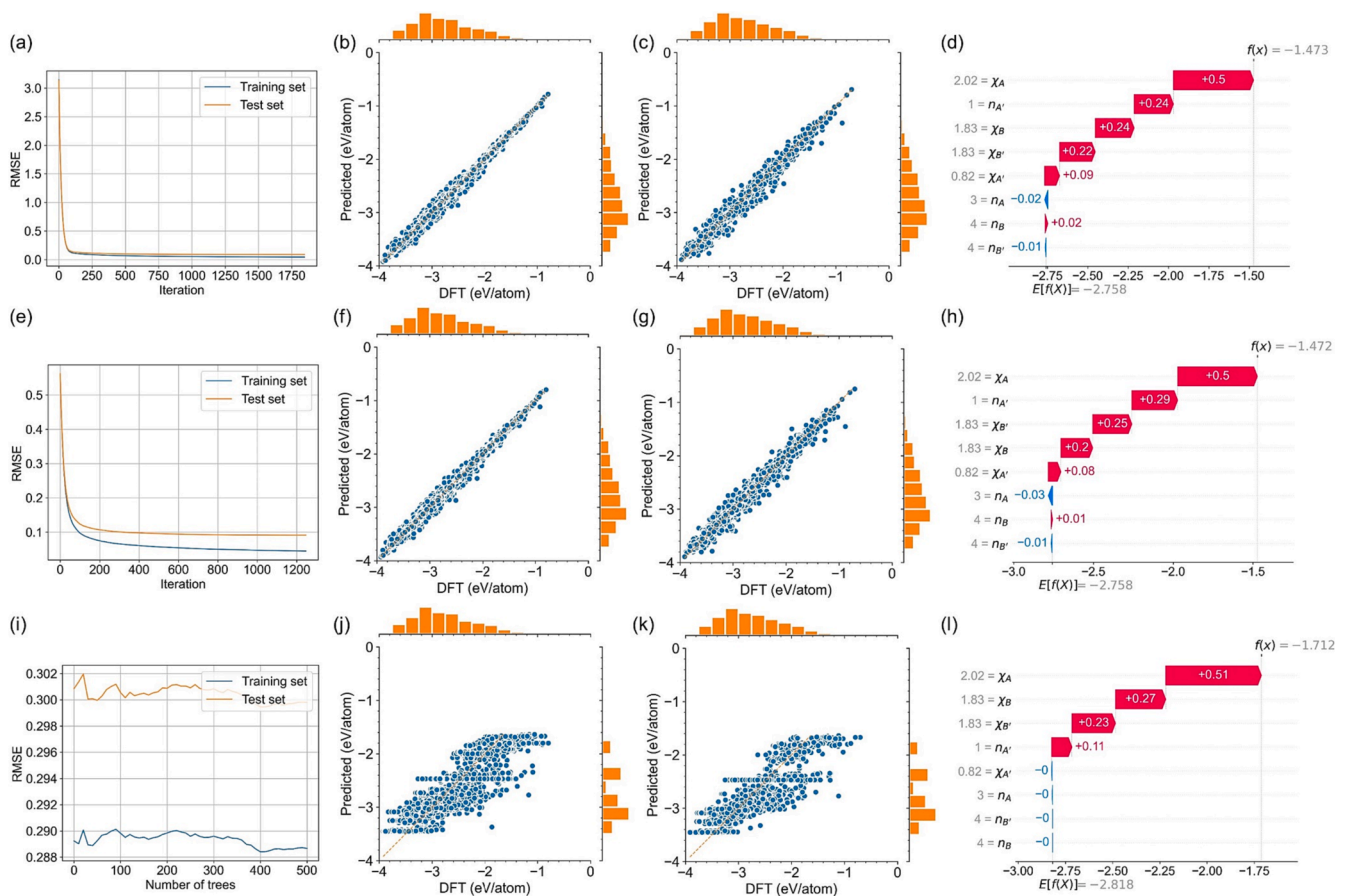


Fig. 2. Evaluation of ML models on formation enthalpy prediction: (a, e, and i) Loss function of XGBoost, LightGBM, and random forest algorithms on training and test datasets, respectively; (b and c) XGBoost predictions vs. DFT calculations on training and test sets; (f and g) LightGBM predictions vs. DFT calculations on training and test sets; (j and k) random forest predictions vs. DFT calculations on training and test sets; (d, h, and l) SHAP analysis of XGBoost, LightGBM, and random forest models, respectively.

respectively, from Fig. 2a and Fig. 2e. Fig. 2i reveals that the optimal number of trees for the random forest is 400. XGBoost performed better on the training set compared to the test set with MAE, RMSE, and R^2 values on the training set (Fig. 2b) being 0.026 eV/atom, 0.043 eV/atom, and 0.99, respectively. Correspondingly, on the test set (Fig. 2c), the values were 0.062 eV/atom, 0.089 eV/atom, and 0.98. The performance of LightGBM was comparable to that of XGBoost, with metrics on the training set (Fig. 2f) being 0.029 eV/atom, 0.045 eV/atom, and 0.99, and on the test set (Fig. 2g) being 0.063 eV/atom, 0.091 eV/atom, and 0.98. The random forest lagged behind both XGBoost and LightGBM. Its predictions showed a more significant deviation from the DFT values, yielding metrics on the training set (Fig. 2j) of 0.224 eV/atom, 0.289 eV/atom, and 0.75, and on the test set (Fig. 2k) of 0.231 eV/atom, 0.300 eV/atom, and 0.74. To enhance the interpretability of our machine learning outcomes, we integrated SHAP analysis (as shown in Fig. 2d, 2h, and 2l). For the 6526 AA'BB'O₆ compositions, across the XGBoost model, the oxidation states of the A' site atoms, as well as the electronegativities of both the A, B, and B' site atoms, played a significant role in influencing the formation enthalpies. This insight provides a critical factor for experimentally synthesizing stable perovskites. However, a clear functional relationship between input features, including oxidation states and electronegativity, and formation enthalpies remains to be established.

Symbolic regression based on genetic programming offers new insights into deciphering the mathematical relationship between the intrinsic attributes of perovskite structures (oxidation states and electronegativity) and material stability (formation enthalpy). The performance of symbolic regression surpasses that of random forest with an MAE, RMSE, and R^2 of 0.207 eV/atom, 0.266 eV/atom, and 0.79, respectively, for the training set (Fig. 3a). For the test set (Fig. 3b), these values are 0.208 eV/atom, 0.270 eV/atom, and 0.79, respectively. Although symbolic regression doesn't outperform XGBoost and LightGBM, the mathematical relationship it derives, namely $E_f = -\left(-\cos(\chi_B^{2.78}) - \cos\left(\frac{1.02\text{tan}(\chi_B)}{\chi_B}\right) - \cos^3(e^{\chi_B} - 0.258) + \text{acosh}(n_A)\right) + \text{asinh}^{-n_B}(\chi_B) \cdot \text{atan}\left(\frac{0.376}{\chi_B^{0.887}}\right) - \sin^3(1.43\chi_A \cos(\chi_A) + 1.14) - 1.54$, offers a clear understanding of the influence of different input features on stability. Based on the expression, χ_B and χ_B , appearing in various complex terms, have a more pronounced effect on the decomposition enthalpy. To further quantify the influence of distinct input features on stability, we incorporated sensitivity analysis. In discerning the influence of input variables ($n_A, n_A', n_B, n_B', \chi_A, \chi_A', \chi_B, \text{ and } \chi_B'$) on the output y , we utilized this analysis. With all other variables held constant at their median values, y was evaluated at the bounds of each variable. Sensitivity was defined as $S_{x_i} = |y(x_i = \text{high}, \text{others at center}) - y(x_i = \text{low}, \text{others at center})|$, where Δy represents the change in output y as x_i varies between its "high" and "low" values, which are its upper and lower limits. The variables n_A to n_B had values ranging from 1 to 7, while χ_A to χ_B ranged from 0.82 to 2.55. The sensitivity analysis indicated a hierarchy of influence: $S_{\chi_B} > S_{n_A} > S_{\chi_A} > S_{\chi_B'}$. This order is similar to the findings from the earlier SHAP

analysis based on the XGBoost model, which indicated $S_{\chi_A} > S_{n_A} > S_{\chi_B} > S_{\chi_B'}$. This methodology underscores the variables with substantial impacts on y , furnishing insights into the system's behavior and spotlighting potential areas warranting closer examination.

We selected the top 500 multinary perovskite oxides based on stability rankings for band gap calculations using the PBE functional. Given the Shockley–Queisser limit [29] and the known tendency of the PBE functional to underestimate the band gap [40,41], we focused on three compounds with band gaps of 0.38 eV, 0.85 eV, and 1.41 eV: DyNdHf₂O₆, CeEuAl₂O₆, and CeSrAl₂O₆. These were further evaluated using the HSE06 functional. The optimized structures of these compounds are illustrated in Fig. S1. From the PDOS, it's evident that the conduction band minimum (CBM) and valence band maximum (VBM) of DyNdHf₂O₆ are primarily contributed by the Nd and Hf orbitals (Fig. 4a). For both CeEuAl₂O₆ (Fig. 4b) and CeSrAl₂O₆ (Fig. 4c), the CBM and VBM mainly arise from the Ce and O orbitals. The atomic orbital contributions to the band edge states, as shown in the band diagrams (Fig. 4d, 4e, and 4f), align with the PDOS observations. The band gaps of DyNdHf₂O₆, CeEuAl₂O₆, and CeSrAl₂O₆ are found to be 0.87 eV, 2.31 eV, and 2.85 eV, respectively. Notably, the HSE06 band gap of CeEuAl₂O₆ is closely aligned with that of 2D perovskites utilized in solar cells [42,43]. Influenced by their band gaps, the average optical absorption in the visible range decreases sequentially for the three compounds, as evident from the optical absorption spectra (Fig. S2). However, only CeEuAl₂O₆ exhibits a direct band gap semiconductor characteristic. The energy difference between direct and indirect transitions for DyNdHf₂O₆ and CeSrAl₂O₆ are 50 meV and 31 meV, respectively. Hence, considering both the direct band gap nature and the magnitude of the band gap, CeEuAl₂O₆ emerges as a promising candidate for photovoltaic applications.

Based on Fermi's golden rule [44] and the Franck-Condon principle [45], reduced wave function overlap leads to decreased non-adiabatic coupling matrix elements, consequently diminishing charge and energy losses during the non-radiative electron-hole recombination process. It is thus essential to enhance the optoelectronic properties of candidate materials by minimizing the charge density overlap between the CBM and VBM. As depicted in Fig. 5 and Fig. S3, the CBM and VBM charge densities for DyNdHf₂O₆ are localized on the Nd and Hf atomic sites respectively, with both exhibiting comparable localization extents. For CeEuAl₂O₆ and CeSrAl₂O₆, the CBM and VBM charge densities are localized at the Ce and O atomic sites, respectively. Given the more even spatial distribution of O atoms compared to Ce atoms within the crystal lattice, the CBM displays a more pronounced degree of localization than the VBM.

To offer a precise description of the variance in charge localization or delocalization, we employed the Inverse Participation Ratio (IPR) [46], which signifies the localization or delocalization degree of Kohn-Sham states. The IPR for a specific Kohn-Sham state k_i is defined as $\text{IPR} = \frac{1}{N} \left(\frac{\sum_i k_i^2}{\sum_i k_i} \right)^2$, with $\text{IPR} \in (0, 1]$. An IPR value of 1 indicates a fully delocalized charge state, while lower IPR values denote increased localization. For the DyNdHf₂O₆, CeEuAl₂O₆, and CeSrAl₂O₆ systems, the IPRs for CBM are 0.193, 0.042, and 0.054 respectively, and for VBM they are 0.182, 0.121, and 0.161. The trend observed in IPR values further substantiates the conclusions from the charge projection analysis. CeEuAl₂O₆, bearing a direct bandgap, not only showcases a suitable bandgap size but also the least electron-hole wave function overlap, implying reduced non-adiabatic coupling. Such attributes render it promising for outstanding photovoltaic device applications.

4. Conclusions

In this study, we harnessed machine learning techniques to forecast the stability of 6526 AA'BB'O₆ perovskite compositions. Notably, both the XGBoost and LightGBM models demonstrated superior performance

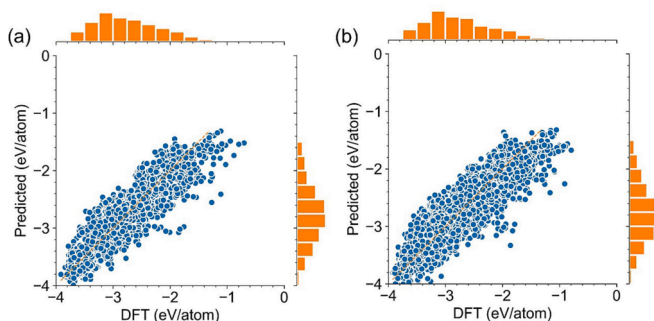


Fig. 3. Symbol regression predictions vs. DFT calculations on (a) training and (b) test sets.

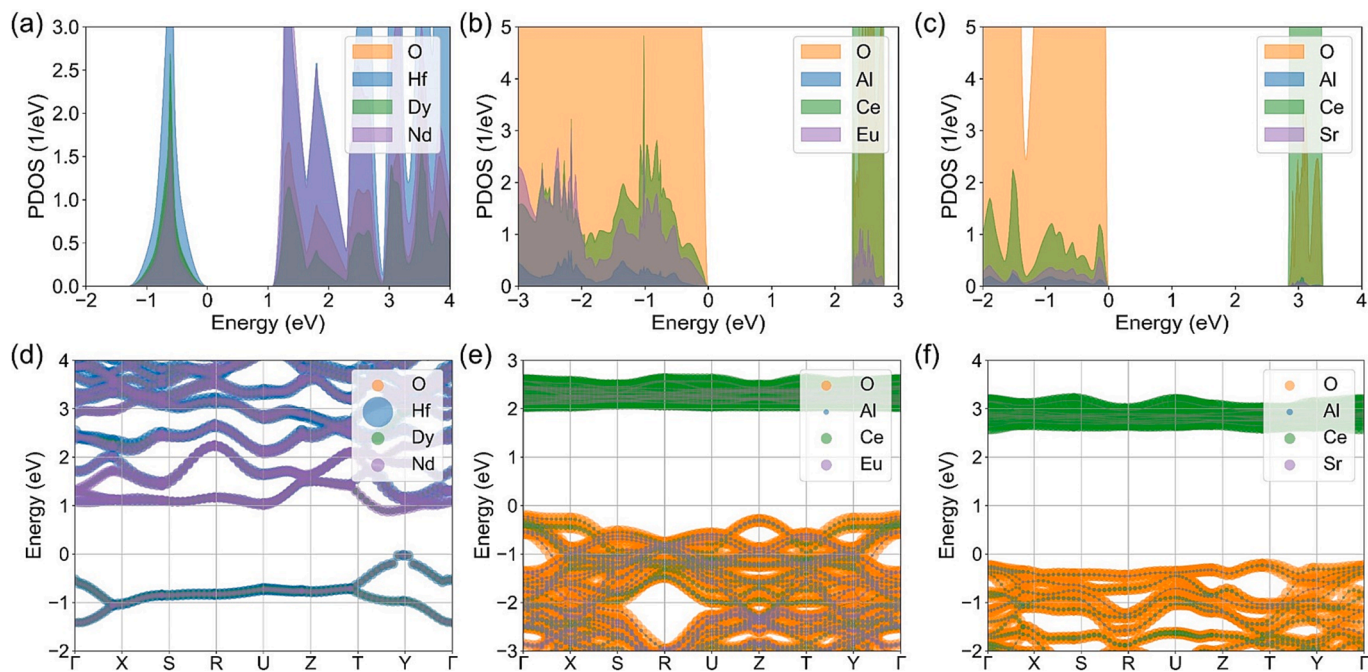


Fig. 4. PDOS at the top panel and band structures at the bottom panel, determined using the HSE06 functional, for (a and d) DyNdHf₂O₆, (b and e) CeEuAl₂O₆, and (c and f) CeSrAl₂O₆ perovskites. The zero energy reference is set to the Fermi level.

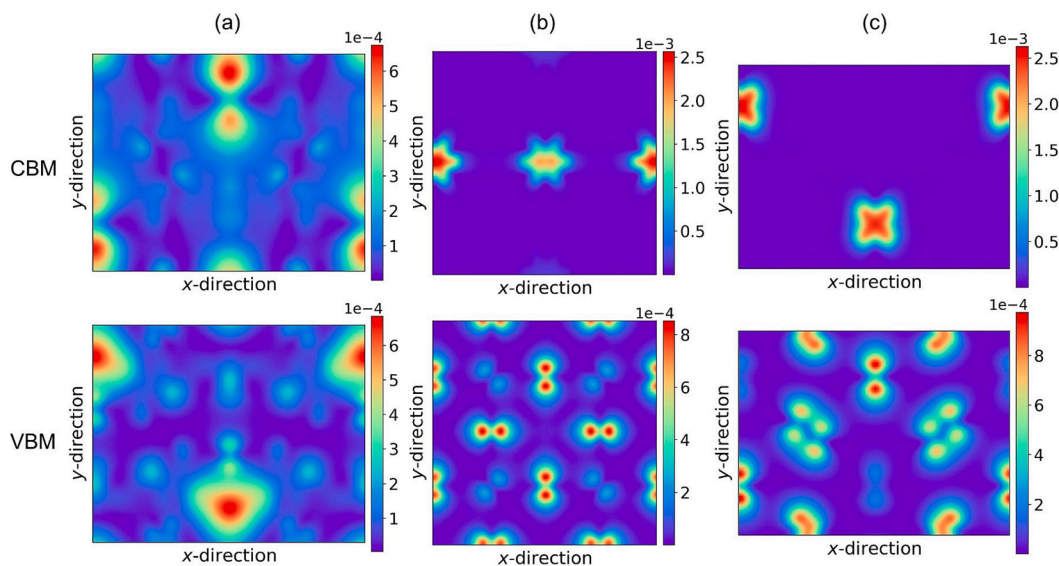


Fig. 5. Charge density projections of CBM and VBM along the z-axes for (a) DyNdHf₂O₆, (b) CeEuAl₂O₆, and (c) CeSrAl₂O₆ perovskites.

over the random forest approach. To deepen the interpretability of our ML models, SHAP analysis was applied. It revealed that parameters like the oxidation state of the A' site atoms, the electronegativity of the A site atoms, and the electronegativity of both B and B' site atoms played pivotal roles in dictating the formation enthalpy. Seeking a transparent, quantitative relationship between these influential input features and formation enthalpy, we integrated a genetic programming-based symbolic regression algorithm. Paired with sensitivity analysis, our results echoed the insights garnered from SHAP, but with an added advantage: the ability to pinpoint precise mathematical correlations. These insights paved the way for the conceptualization and design of multinary perovskite oxides boasting enhanced stability, thus bolstering their theoretical foundations. Venturing into the realm of photovoltaics, we earmarked the top 500 materials in stability rankings for PBE bandgap evaluations. A deeper dive, leveraging the HSE06 functional, spotlighted

DyNdHf₂O₆, CeEuAl₂O₆, and CeSrAl₂O₆ as potential game-changers. Among these, CeEuAl₂O₆ stands out, positioning itself as a promising contender in the photovoltaic material lineup, owing to its direct bandgap characteristics, a comparable bandgap to feasible 2D perovskite materials, and diminished non-adiabatic coupling effects. This research casts a theoretical spotlight, guiding the application of perovskite oxides in photovoltaics. Moving forward, our research trajectory is geared toward refining the precision of symbolic regression. Our ambition is to achieve a model that blends high accuracy with rich interpretability, a combination essential for pioneering materials of unmatched efficacy.

Data availability

The data set, along with detailed evaluations of input features using

XGBoost, optimal symbolic regression hyperparameters, and analyses of perovskite geometries, optical absorption spectra, and charge densities, is available in the [Supplementary Data](#).

CRedit authorship contribution statement

Zhaosheng Zhang: Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Yingjie Zhang:** Data curation, Validation. **Sijia Liu:** Data curation, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (22103021), the Natural Science Foundation of Hebei Province (B2020201070), and the Advanced Talents Incubation Program of Hebei University (521000981344). DFT calculations were performed at the High-Performance Computing Center of Hebei University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.commatsci.2024.112889>.

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Supporting Information for

**Integrative approach of machine learning and symbolic regression for
stability prediction of multicomponent perovskite oxides and high-
throughput screening**

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Table S1 Performance of various input features on the test set evaluated by R^2 using the XGBoost algorithm.

input features	R^2
oxidation state and ionic radii	0.96
oxidation state and electronegativity	0.98
ionic radii and electronegativity	0.97
oxidation state, ionic radii, and electronegativity	0.97

Table S2 Optimal hyperparameter settings for the symbolic regression algorithm based on the PySR code.

hyperparameter	value
binary_operators	['+', '-', '*', '/', '^']
unary_operators	['neg', 'square', 'cube', 'exp', 'abs', 'log', 'log10', 'sqrt', 'sin', 'cos', 'tan', 'atan', 'asinh', 'acosh', 'inv(x)=1/x']
extra_sympy_mappings	{"inv": lambda x: 1 / x}
niterations	80
populations	30
population_size	100
ncyclesperiteration	1100

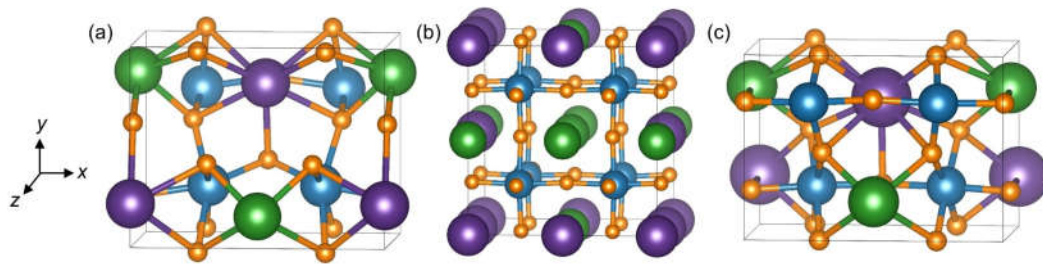


Fig. S1. Illustration of simulation cells depicting the geometries of (a) $\text{DyNdHf}_2\text{O}_6$, (b) $\text{CeEuAl}_2\text{O}_6$, and (c) $\text{CeSrAl}_2\text{O}_6$ perovskites.

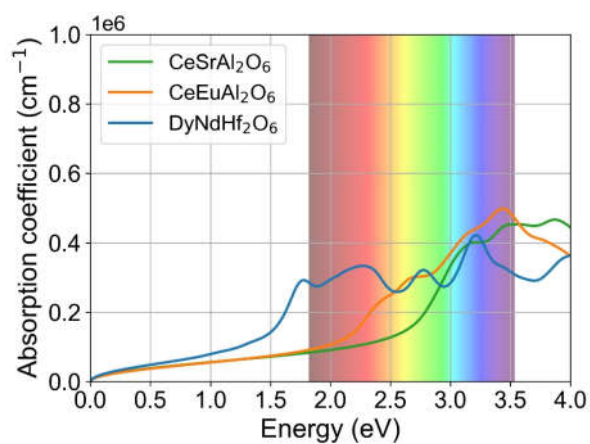


Fig. S2. Optical absorption spectra of $\text{DyNdHf}_2\text{O}_6$, $\text{CeEuAl}_2\text{O}_6$, and $\text{CeSrAl}_2\text{O}_6$ perovskites using the HSE06 methodology.

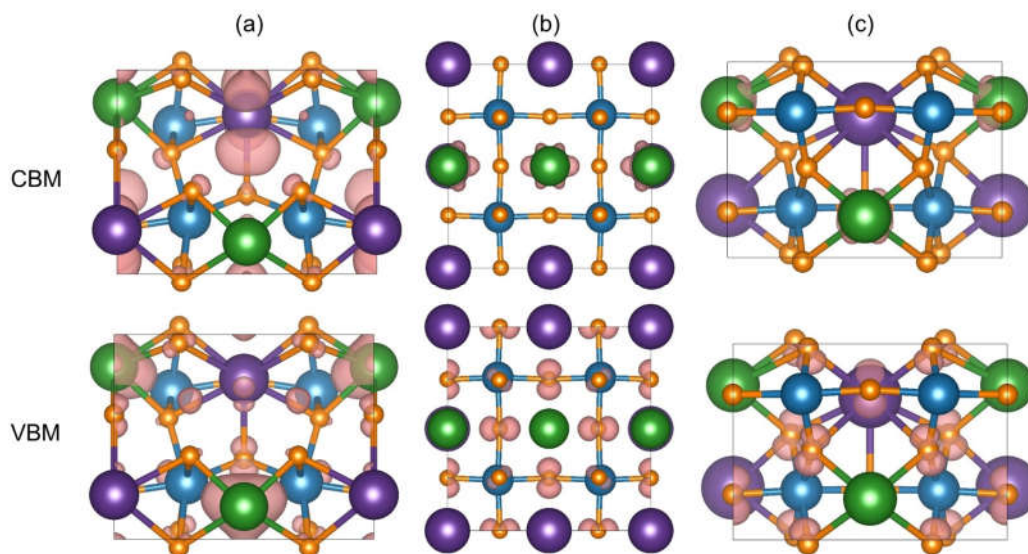


Fig. S3. Charge densities of CBM and VBM for (a) $\text{DyNdHf}_2\text{O}_6$, (b) $\text{CeEuAl}_2\text{O}_6$, and (c) $\text{CeSrAl}_2\text{O}_6$ perovskites, corresponding to Fig. 5 in the main text.